# Rishi More

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## Summary

ML engineer building **risk-controlled LLM inference**, **retrieval systems**, and **production-grade ML pipelines**. Co-first author of an **ICML 2026** paper on distribution-free risk control for reducing test-time compute under calibrated error guarantees. Experienced in turning research prototypes into low-latency ML pipelines, evaluation harnesses, and production-facing systems.

## Experience

**JHU Data Science and AI Institute (DSAI)** — **May 2025 – Present**
*Graduate Research Assistant* (Advisors: Prof. Daniel Khashabi, Prof. Eric Nalisnick) — *Baltimore, MD*

- **Co-first author** of *Conformal Thinking* (**ICML 2026**): distribution-free risk control that sets dual confidence thresholds for adaptive reasoning, cutting LLM inference cost while bounding error at a user-specified risk level.
- Built a vLLM-based calibration/evaluation pipeline across **4** reasoning models and **4** benchmarks (**2.7K+** problems), maintained target error rates under dataset/length shift using finite-sample (UCB) calibration.
- Designed dual-threshold stopping rules, including a parametric lower threshold and ensembled halt policy, reducing reasoning tokens by **12%** at matched accuracy with reusable evaluation tooling.

**JHU Center for Language and Speech Processing (CLSP)** — **January 2025 – December 2025**
*Graduate Researcher* (Advisor: Prof. Jason Eisner) — *Baltimore, MD*

- Built a temporal-aware RAG pipeline over **10K+** non-stationary documents using time-decay weighting, metadata-aware indexing, and temporally scoped retrieval.
- Improved retrieval precision by **23%** over dense baselines in high-volatility knowledge domains by prioritizing time-relevant evidence.
- Reduced false retrievals by **31%** using reference-free consistency scoring to detect stale or hallucination-prone retrieved context.

**Mastek** — **November 2022 – July 2023**
*Machine Learning Engineer Intern* — *Mumbai, India*

- Built and deployed an on-prem **edge** voice authentication system at **<100ms p99** latency and **96.6%** accuracy, using a lightweight Siamese mel-spectrogram model for retraining-free enrollment under edge-compute limits.
- Engineered a TensorFlow pipeline over **1M+** voice samples with spectral-gating denoising for robust feature extraction.
- Improved authentication reliability **20%** in high-noise production environments via preprocessing and feature-extraction upgrades.

## Projects

**Post-Training Evaluation & Reward-Hacking Detector** | *Python, PyTorch, DPO, LLM-as-Judge, Calibration* | GitHub

- Built an evaluation harness for **6** post-training failure modes—reward hacking, sycophancy, verbosity bias, false confidence, safety drift, and distribution shift across **4** Qwen3-8B variants: Base, SFT, DPO, and Instruct.
- Evaluated **142** adversarial prompts with debiased pairwise LLM judging, position-swapped comparisons, and log-prob calibration, flagging **10%** of DPO preference wins that degraded factuality.
- Diagnosed optimization-stage regressions using ECE, Brier score, AUROC, and bootstrap confidence intervals, finding SFT raised harmful compliance to **12.5%** and DPO increased sycophancy from **34%** to **45%**.

**Asymmetric Relation Dynamics in Language Agents** | *Python, LLM Agents, LoRA, asyncio, BM25* | GitHub

- Built an async multi-agent training system (Tinker API): a frozen **Qwen3-235B** caregiver teaching LoRA-adapted **Qwen3-8B** agents via a four-module memory and salience-gated BM25 retrieval.
- Reached **100%** training success with ∼**25%** fewer turns (6.1 vs 8.4) over Solo/Peer baselines across **4** conditions × **3** seeds.
- Isolated scaffolding dependency as the transfer bottleneck via ablations; fully reproducible via a single-command CLI.

**Decode Roofline: Kernel-Level Profiling & Fused INT4 Dequant-GEMV** | *CUDA C++, Nsight, PyTorch, vLLM* | GitHub

- Profiled batch-1 LLM decode (**Qwen2.5-1.5B**) at the CUDA-kernel level on a consumer GPU, using Nsight to attribute ∼**81%** of kernel time to weight-streaming GEMVs pinned at ∼**95%** of the measured ∼**250 GB/s** memory roofline.
- Wrote a correctness-gated fused **INT4** dequant + GEMV CUDA kernel hitting **86%** of peak bandwidth (∼**8.7**× fewer bytes than a two-op baseline), winning **2.38**× at batch-1 then ceding to FP16 GEMM by batch-2 as arithmetic intensity rises.

## Education

**Johns Hopkins University** — **May 2026**
*M.S.E. in Computer Science* — *GPA: 4.0/4.0*

- **Coursework:** Theory of Replicable ML, Human-in-the-Loop ML, Non-Stationary Environments.

**University of Mumbai** — **July 2024**
*B.Tech. in Computer Science, Honors in AI/ML* — *CGPA: 9.85/10*

## Technical Skills

**Languages**: Python, C++, Java, C
**ML/LLM Systems**: PyTorch, TensorFlow, Hugging Face, LoRA, DPO, RAG, calibration, conformal/risk control, Scikit-learn
**Inference & MLOps**: vLLM, CUDA C++, Nsight Compute/Systems, Docker, Kubernetes, AWS, Apache Airflow, Apache Spark
**Data**: SQL, PostgreSQL, MongoDB, Pandas, NumPy